Recurrence Relations

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Recurrence relations

Help me understand

What is a recurrence relation?

- A **recurrence relation** describes each term in a sequence as a function of the previous term – ie $u_{n+1} = f(u_n)$
- Along with the first term of the sequence, this allows you to generate the sequence term by term

e.g. FIND THE FIRST FOUR TERMS OF THE SEQUENCE DEFINED BY $u_{n+1} = u_n + 3$, $u_1 = 2$

$u_1 = 2$ $u_2 = u_1 + 3 = 2 + 3 = 5$

$u_3 = u_2 + 3 = 5 + 3 = 8$

$u_4 = u_3 + 3 = 8 + 3 = 11$

THE FIRST FOUR TERMS ARE 2, 5, 8, 11

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- Both **arithmetic sequences** and **geometric sequences** can be defined using recurrence relations
  - **Arithmetic** can be defined by $u_{n+1} = u_n + d, u_1 = a$
  - **Geometric** can be defined by $u_{n+1} = u_n \times r, u_1 = a$

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

(i.e. ARITHMETIC SEQUENCE WITH FIRST TERM 9 AND COMMON DIFFERENCE −5)

$$u_{n+1} = 2u_n, \quad u_1 = 3$$

THIS IS THE SEQUENCE 3, 6, 12, 24, 48, ...
(i.e. GEOMETRIC SEQUENCE WITH FIRST TERM 3 AND COMMON RATIO 2)

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- However, you can also define sequences that are neither arithmetic nor geometric

$$u_{n+1} = 2u_n + 3, \quad u_1 = 1$$

THIS IS THE SEQUENCE 1, 5, 13, 29, 61, ...
(DON'T CONFUSE THIS WITH ARITHMETIC SEQUENCE 5, 7, 9, 11, ... DEFINED BY $u_n = 2n + 3$)

$$u_{n+1} = (u_n)^2, \quad u_1 = 2$$

THIS IS THE SEQUENCE 2, 4, 16, 256, 65 536, ...

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Examiner Tips and Tricks

- For arithmetic or geometric sequences defined by recurrence relations, you can sum the terms using the **arithmetic series** and **geometric series** formulae.
- To sum up the terms of other sequences, you may have to think about the series and find a clever trick.

Worked Example

$$u_1 = 2$$
$$u_{n+1} = u_n^2 - 3$$

Find $\sum_{r=1}^{101} u_r$

$u_1 = 2 \quad u_2 = 2^2 - 3 = 1 \quad u_3 = 1^2 - 3 = -2$

$u_4 = (-2)^2 - 3 = 1 \quad u_5 = 1^2 - 3 = -2$

WORK OUT THE TERMS AND LOOK FOR A PATTERN

SO THE SEQUENCE IS

$2, 1, -2, 1, -2, 1, -2, \ldots$

IN THE FIRST 101 TERMS THERE WILL BE

ONE 2 $\quad 1 \cdot 2 = 2$

FIFTY 1's $\quad 50 \cdot 1 = 50$

FIFTY −2's $\quad 50 \cdot (-2) = -100$

$2 + 50 + (-100) = -48$

SO $\sum_{r=1}^{101} u_r = -48$

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